

Sai Kiran Belana

saikiran.belana@uconn.edu | +1-959-995-3083

 belanasaikiran.github.io |  linkedin.com/in/belanasaikiran/ |  github.com/belanasaikiran

EDUCATION

University of Connecticut

January 2024 - December 2025

Masters in Computer Science Engineering

GPA: 3.6/4.0

Coursework: Operating Systems, Advanced Computer Architecture, Algorithms & Complexities, Computer Networks & Data Communications, Computer Security, Cyber Security, Big Data Analytics

SKILLS

Programming Languages: JavaScript, Python, Dart, Bash, C, C++, PHP, HTML5, CSS, SQL, Golang

Frameworks: NodeJs, ReactJs, OpenCL, Django, Flutter, SQL, MongoDB, Laravel, RestAPI, Dear ImGui

Software Tools: Linux, Version Control, Docker, Kubernetes, Git, Slurm Workload Manager, Vagrant, Yocto Project, Figma, Adobe, Nginx, Markdown

Technologies: Microsoft Azure, Amazon Web Services, Google Cloud, On-Premise, Distributed Computing

PROFESSIONAL EXPERIENCE

Student Developer - ITS, University of Connecticut

February 2025 - Present

- Developed additional features for WordPress plugins within the Content Management System (Aurora), deploying changes to test environments and verifying the functionality of plugins and applications.
- Updated WordPress plugins based on new specifications, deploying them to test servers using OpenShift & bitbucket pipelines.
- Optimized the University Classrooms system by consolidating five separate classroom sites into a single unified platform, reducing maintenance costs by 40% and improving data synchronization across all sites.

Software Engineer, LightSpeed Photonics

March 2021 - January 2024

- Distributed the computation of scalable tile-based matrix multiplication across multiple Intel Arria 10 FPGAs, benchmarking kernel execution performance for Xeon CPUs using OpenCL.
- Generated Petalinux OS images for Xilinx FPGA MPSoC boards, created bash scripts for image generation, and developed GUIs to demonstrate video analytics on custom EdgeAI Acceleration Hardware.
- Developed and deployed Intel FPGA-based Docker images to an on-premise Kubernetes cluster using webhooks, successfully integrating them with Azure Kubernetes Service.
- Created a web application platform using Slurm Workload Manager to deploy applications on FPGAs running on an HPC Cluster, providing remote access and developer utilities.
- Built the company's portfolio website, including hosting, cloud databases, CDNs, and email services, using ReactJS, Azure Functions, Azure CDNs, and SendGrid API Integration.
- Contributed to the full software development lifecycle and collaborated with teams to build a user-friendly real-time web app for running FPGA applications on custom hardware.

PROJECTS

OS/161 Kernel System Calls Implementation and Testing

Tools/Languages: [Operating Systems, C, Linux, GCC, Makefile]

- Created multi-threaded applications with thread lock handling to optimize concurrency and resource utilization.
- Implemented new system calls, locks, and semaphores in OS/161 to manage process synchronization and prevent deadlocks, improving system stability.
- Developed user-level programs to test the functionality of these system calls, ensured proper execution through kernel modifications, and successfully validated the results by printing them to the console.

File System Management - Operating Systems

Tools/Languages: [Operating Systems, C++, Linux, G++, Makefile]

- Developed a file system GUI using Dear ImGui, enabling users to manage files, directories, monitor disk usage, and perform file modifications.
- Implemented core functionality in C++, leveraging OpenGL for efficient rendering of file system operations and UI components.
- Enhanced user experience with real-time updates and smooth interactions for file management tasks and system monitoring.

Pipelined RISC-V Simulator in C

Tools/Languages: [C, GCC, Assembly]

- Developed a functional 5-stage pipelined RISC-V processor simulator, supporting all instruction types and enhancing simulation accuracy.

- Implemented dynamic branch predictors, pipeline interlocks, and forwarding techniques to handle data and structural hazards, ensuring accurate simulation results under various unit tests.

Car Make & Model Detection using Xception Model (Transfer Learning) 🔗

Tools/Languages: [Python, Keras, TensorFlow, Xception, Stanford Car Dataset]

- Developed a robust deep learning model to classify car make and model from the Stanford Car Dataset (196 classes, 16,185 images) using the Xception architecture for transfer learning.
- Achieved 99.69% accuracy and 0.58% loss on unseen data, outperforming all existing Kaggle implementations and demonstrating high scalability for real-world applications like traffic monitoring and accident analysis.
- Designed end-to-end workflows, including image preprocessing, training optimization, and video processing scripts, with well-documented Jupyter Notebooks for reproducibility.

IPv4 Router with OpenFlow and POX Controller 🔗

Tools/Languages: [Python, OpenFlow, Mininet, iperf]

- Designed and implemented a network routing interface to enable communication between devices on different subnets.
- Configured firewalls for specific subnets to simulate a production network environment on Mininet.
- Developed custom network rules to control and manage traffic flow across the network.
- Conducted thorough testing of network packet flows using Linux tools such as iperf, ping, and dpctl.

Cross-Platform Notification Synchronization Application using Sockets and Bluetooth Low Energy 🔗

Tools/Languages: [Flutter, BLE]

- Working on a Flutter-based application to develop native cross-platform software that synchronizes real-time messages between desktop and mobile devices.
- Implementing features to synchronize mobile notifications, real-time calls, and clipboard sharing with desktop systems.

Web-Based Terminal using WebSockets 🔗

Tools/Languages: [ReactJS, NodeJS, ExpressJS, xTerm.js, Socket.io, SSH2]

- Built a web terminal interface to enable secure remote access to cloud systems.
- Utilized xTerm.js for terminal emulation and Socket.io for real-time communication.
- Deployed the application on Vercel, ensuring device-agnostic accessibility and seamless performance.

CERTIFICATIONS

Docker Mastery: with Kubernetes + Swarm from a Docker Captain - Bret Fisher

Issued By: Udemy

The Complete Web Development Bootcamp - Angela Yu

Issued By: Udemy

Intel OneAPI Hackathon - 2023

Issued By: Intel Corporation